

Total no. of Pages: 1

Roll no.

Degree: B-Tech.

Semester 3rd

MID-SEMESTER EXAMINATION, June, 2026

Course Title: Probability Theory and Random Process

Course Code: Sem: ECECC303/EIECC303/VTECC303/ECECC06

Duration: 1:30 Hours

Max. Marks: 20

Note: - Attempt all questions in the given order only. Missing data/information (if any), may be suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO																				
1a	Define a Random Variable. Differentiate between discrete and continuous random variables.	1	C01																				
1b	Let X be a continuous random variable with the following distribution: $f(x) = \begin{cases} kx, & \text{if } 0 \leq x \leq 5 \\ 0, & \text{elsewhere} \end{cases}$ (a) Evaluate k . (b) Find $P(1 \leq X \leq 3)$. (c) Find $P(X \leq 3)$.	3	C02																				
2a	Distinguish between independent and uncorrelated random variables.	1	C01																				
2b	A fair coin is tossed four times. Let X denote the number of heads occurring. Find: (a) the distribution of X , (b) $E[X]$, (c) the probability graph of X .	3	C02																				
3a	If $E[X] = 3$, $E[Y] = 2$ and $E[XY] = 8$, $Cov(X, Y)$.	1	C02																				
3b	Let X and Y be random variables with joint distribution given below: <table border="1" style="margin: 10px auto;"> <tr> <td>$X \backslash Y$</td><td>-3</td><td>2</td><td>4</td><td>Sum</td></tr> <tr> <td>1</td><td>0.1</td><td>0.2</td><td>0.2</td><td>0.5</td></tr> <tr> <td>3</td><td>0.3</td><td>0.1</td><td>0.1</td><td>0.5</td></tr> <tr> <td>Sum</td><td>0.4</td><td>0.3</td><td>0.3</td><td></td></tr> </table> Find: (a) the distributions of X and Y , (b) $Cov(X, Y)$, (c) $\rho(X, Y)$	$X \backslash Y$	-3	2	4	Sum	1	0.1	0.2	0.2	0.5	3	0.3	0.1	0.1	0.5	Sum	0.4	0.3	0.3		3	C02
$X \backslash Y$	-3	2	4	Sum																			
1	0.1	0.2	0.2	0.5																			
3	0.3	0.1	0.1	0.5																			
Sum	0.4	0.3	0.3																				
4a	If $X \sim N(\mu, \sigma^2)$ with $\mu = 20$ and $\sigma = 4$, write the expression for its PDF.	1	C02																				
4b	A fair coin is tossed 100 times. Find the probability P that the heads occur: (a) exactly 60 times, (b) between 48 and 53 times exclusive (c) less than 45 times.	3	C02																				
5a	Prove $Var(aX + b) = a^2 Var(X)$	1	C01																				
5b	Let X be the binomial random variable $B(n, p)$. Show that (i) $\mu = E[x] = np$ (ii) $Var(X) = npq$.	3	C02																				